# Critical Periods for a First Language (L1)

# Critical periods vs. “sensitive” or “optimal periods”:

**Characteristics of critical periods**

- discrete, "programmed" onset and offset
- change in response to the environment (learning) is strictly limited to this period; no sensitivity to the environment before or after.

**The notion of sensitive or “optimal” periods**

- learning is possible throughout the life-span
- but learning is *optimal during particular windows*
- *these windows might have a gradual onset/offset, which can itself be affected by experience.*

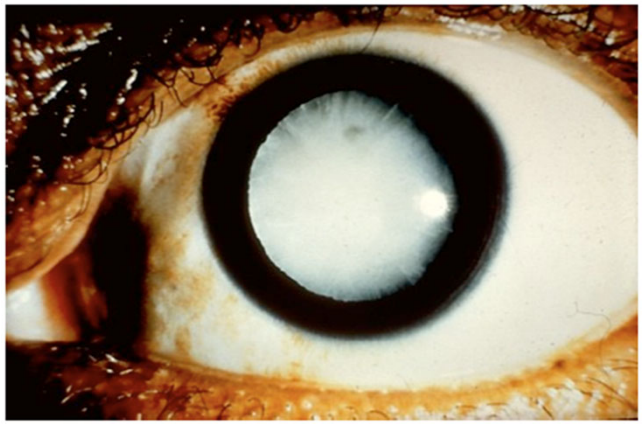
First, we’ll look at a classic “critical period” in vision, and see that even there the term “sensitive period” might be more apt.

Then we’ll look at some of the evidence for critical periods in language learning.

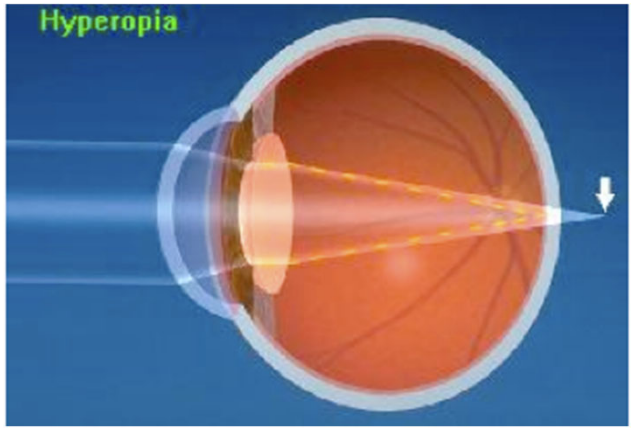
**Amblyopia:** poor vision in one eye despite normal peripheral function



strabismus



cataract

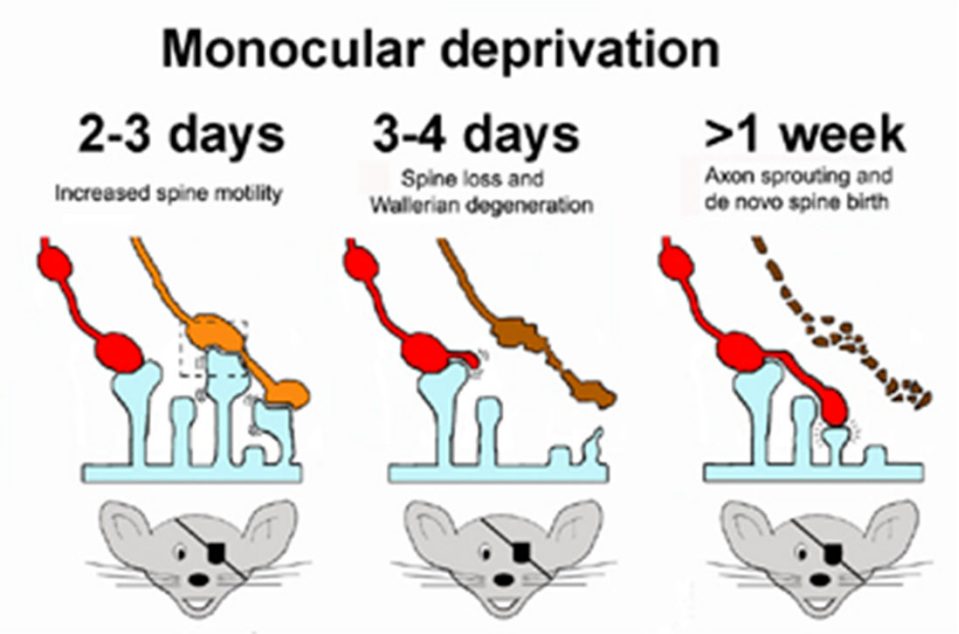



anisometropia

Periods of deprivation from normal visual input caused by these conditions can result in poor vision long after the conditions themselves have been remediated.

*Images courtesy Dennis Levi*

# Amblyopia occurs because of **competitive dynamics** between **cortical inputs**




Abnormal vision in one eye can lead to “colonization” of cortical representation of that eye by the other.

When normal function is restored to the eye, it remains at a competitive disadvantage, and normal cortical responses cannot be restored by normal experience.

# Treatments for amblyopia are based on competition

One of the most effective treatments for amblyopia involves intermittently patching the "good" eye.

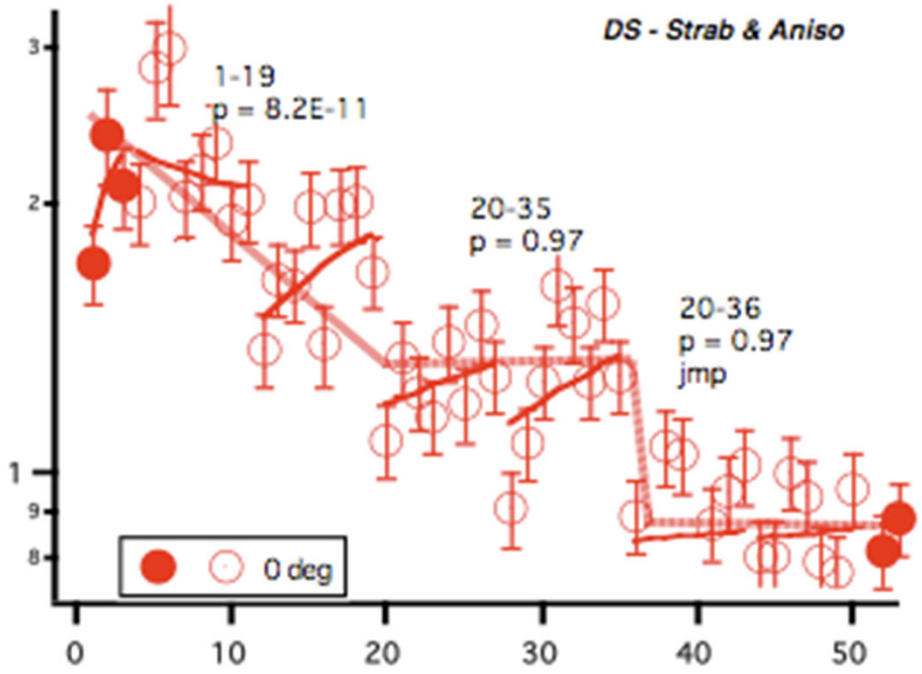
This restores the competitive balance between the eyes and permits the brain's own plasticity to restore normal function.

This treatment is most effective early in development, however, and may be ineffective by late childhood (10y).



*Image courtesy Terri Lewis*

# Human adult plasticity after the critical period

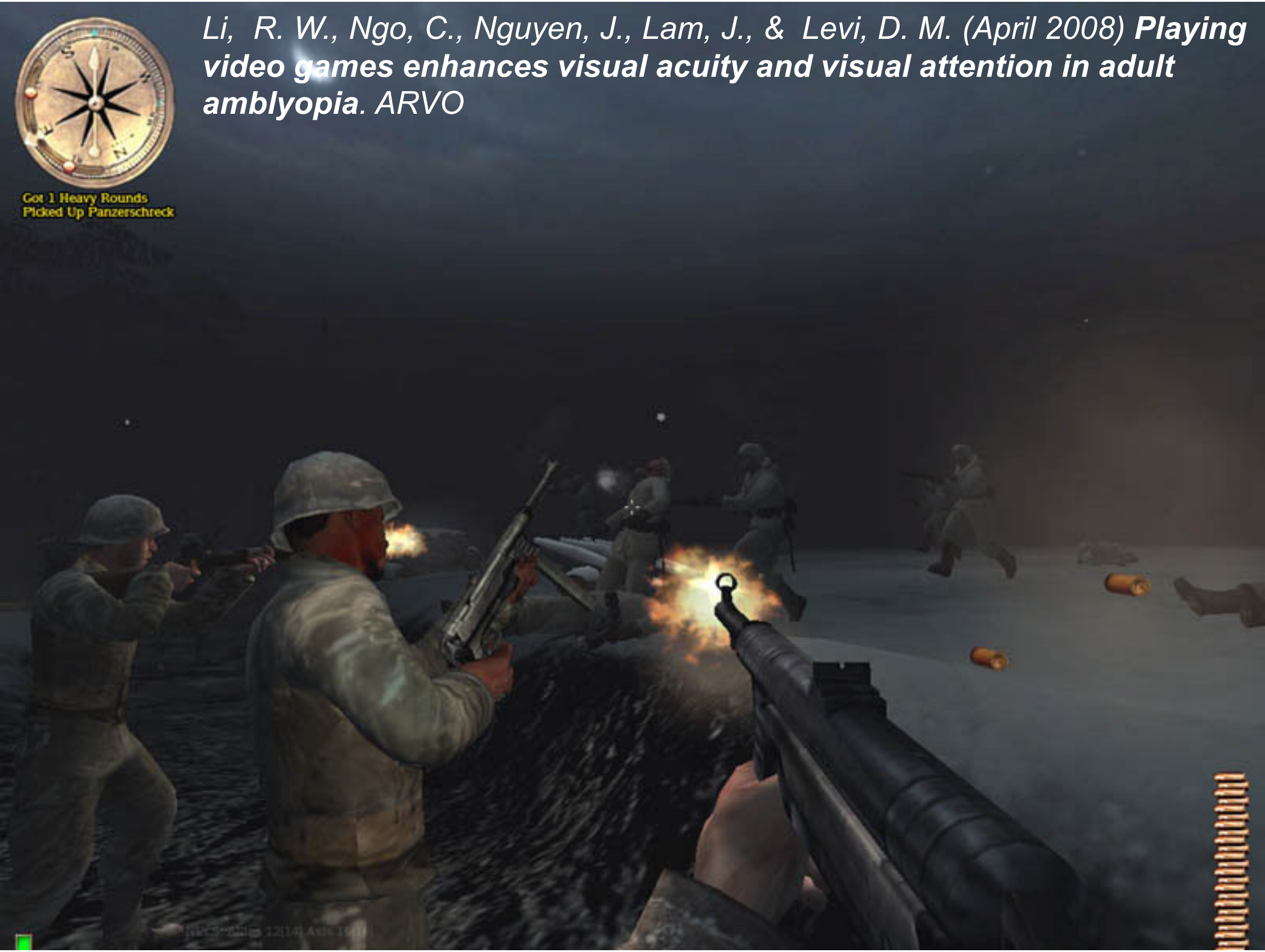



Kilotrials of training

Experiments with adults have begun to demonstrate that plasticity is not abolished entirely, even well into adulthood (some of these patients are in their 50's).

Long-term, intensive training on specific psychophysical tasks in the affected eye can lead to improvements on both task performance and general visual function.

Levi & Polat (1996, *PNAS)*




*Li, R. W., Ngo, C., Nguyen, J., Lam, J., & Levi, D. M. (April 2008)* ***Playing video games enhances visual acuity and visual attention in adult amblyopia****. ARVO*

# Interim summary

Amblyopia is a classic example of a sensitive period:

- peripheral disturbance in early experience can lead to life-long disruption in visual function

- the long-term loss of function is the sequela of a temporary deficit during experience-dependent organization of the brain

- treatment is *most* effective during a limited developmental window

However, new evidence suggests that the *right kind of experience* can reverse defecits late into adulthood.

# How can we relate this to language?

1. Lenneberg's early attempts to synthesize developmental neurology with the notion of a "language acquisition device"

2. Studies of early deprivation: Data from neglected children and Deaf children with limited input to sign.

# Lenneberg's (1967) *Biological Foundations of Language*

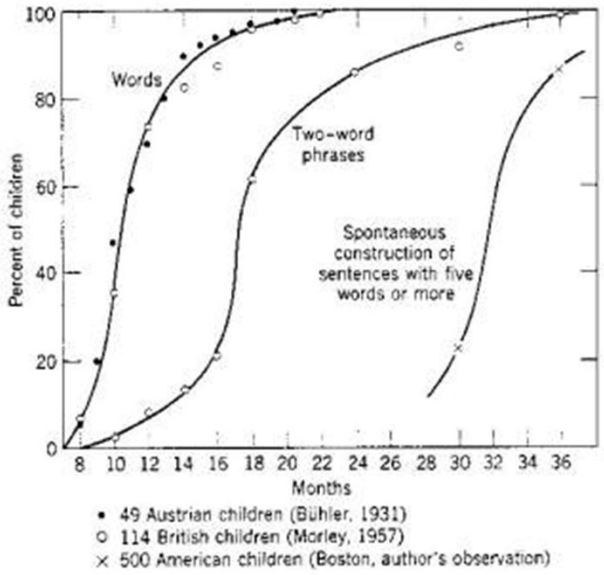



FIG. 4.1. Emergence of various developmental milestones in the acquisit language.

Lenneberg was attempting to synthesize contemporary linguistics with what was known about brain development.

Studies of language development were starting to show a fairly robust and stage-like pattern of language development.

The idea that language might have maturational milestones was seen as consistent with the unfolding of a biological program.

TABLE 4.1 (*Continued*)

| | | |
|---|---|---|
| 8 months | Stands holding on; grasps with thumb apposition; picks up pellet with thumb and finger tips | Reduplication (or more continuous repetitions) becomes frequent; intonation patterns become distinct; utterances can signal emphasis and emotions |
| 10 months | Creeps efficiently; takes side-steps, holding on; pulls to standing position | Vocalizations are mixed with sound-play such as gurgling or bubble-blowing; appears to wish to imitate sounds, but the imitations are never quite successful; beginning to differentiate between words heard by making differential adjustment |
| 12 months | Walks when held by one hand; walks on feet and hands—knees in air; mouthing of objects almost stopped; seats self on floor | Identical sound sequences are replicated with higher relative frequency of occurrence and words (mamma or dadda) are emerging; definite signs of understanding some words and simple commands (show me your eyes) |
| 18 months | Grasp, prehension and release fully developed; gait stiff, propulsive and precipitated; sits on child's chair with only fair aim; creeps downstairs backward; has difficulty building tower of 3 cubes | Has a definite repertoire of words—more than three, but less than fifty; still much babbling but now of several syllables with intricate intonation pattern; no attempt at communicating information and no frustration for not being understood; words may include items such as thank you or come here, but there is little ability to join any of the lexical items into spontaneous two-item phrases; understanding is progressing rapidly |

# Data on typical brain development

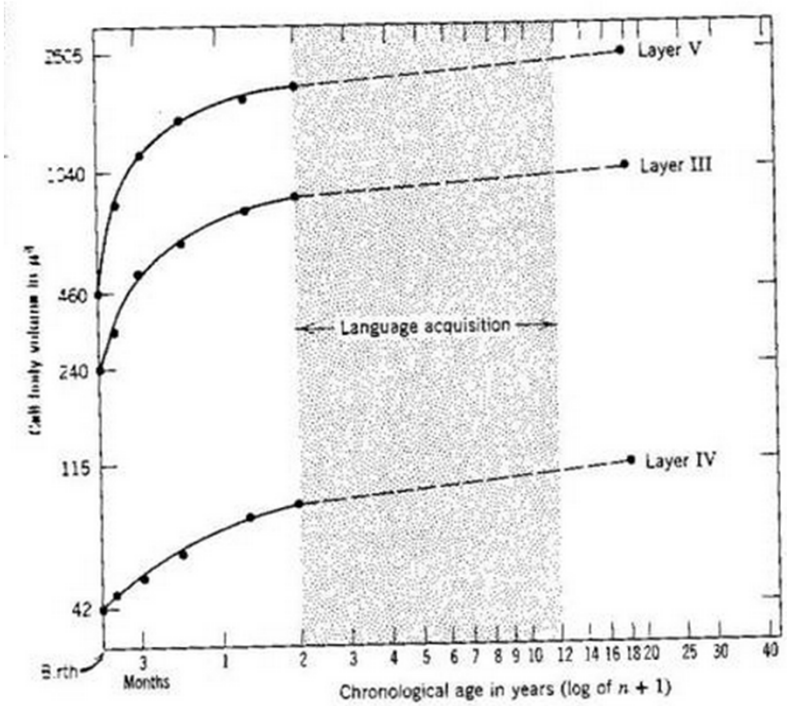



FIG. 4.5. Neurons in the cerebral cortex grow in volume with age. The broken lines between age two and eighteen indicate that no measurements were made on brains of children who died between these ages. (Redrawn from Schadé and van Groenigen, 1961.)

The data on typical brain development were sparse, because they were almost entirely based on post-mortem analyses of tissue.

(See, e.g., the figure depicting neuron size, for which no data were available between ages 2 and 18.)

Nonetheless, Lenneberg observed that abrupt changes in things like neuron size and packing density before age two might be permissive for the beginning of language acquisition.

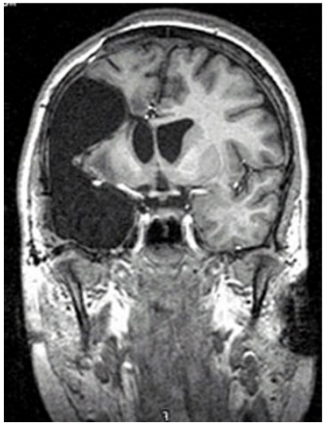
That is, the critical period seems to *open* after this initial period of rapid development.

# More extensive data were available from studies of recovery from stroke.

TABLE 4.7. *Hemispherectomy*

| Lesions Acquired | Hemisphere Operated On | Speech Not Affected or Improved Postoperatively | Permanent Aphasia |
|---|---|---|---|
| Before teens* | Left | 49 | 3 (had aphasia before operation) |
| | Right | 38 | 5 (had aphasia before operation) |
| During puberty† (single case) | Left | Slow improvement | Some residue to end of life (27 months postop.) |
| Adult* | Left | None | 6 (1 had aphasia before operation) |
| | Right | 25 | None |

* Based on Basser, 1962.
† Hillier, 1954 and personal communication.



Aphasia is surprisingly rare in children with brain lesions.

In extreme cases, a *hemispherectomy* was performed (typically to relieve seizures of people with epilepsy).

Children in whom the left hemisphere is *removed* nonetheless seem to develop typical language abilities.

Adults do not recover language after this procedure.

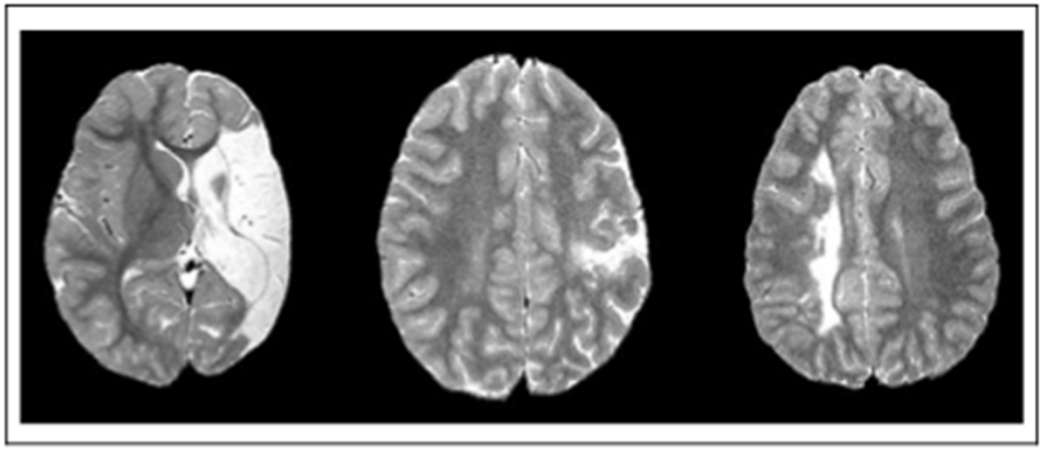
This (among other things) led Lenneberg to hypothesize that a key factor to *closing* the critical period was the development of adult-like *cerebral dominance.*

# As always, things are more complicated.

Systematic studies of children with perinatal (prenatal to 28d) strokes show that outcomes are clearly much better than for adults with similar injuries, BUT:

1. Early language deficits *in both right and left hemisphere patients.* (Some deficits worse for children with RH damage.)

2. Language outcomes generally in low-normal range, suggesting a persistent *delay* rather than a catching up.



# Effects of early deprivation: Genie

**1.** CASE HISTORY. Genie was born in April 1957. When we first encountered her, she was 13 years and 7 months old—a painfully thin child who appeared six or seven years old. When hospitalized for malnutrition, Genie could not stand erect or chew food; she was not toilet trained; and she did not speak, cry, or produce any vocal sounds. The reconstruction of her previous life presents a bizarre and inhuman story. From the age of 20 months, Genie had been confined to a small room under conditions of apparently increasing physical restraint. In this room she received minimal care from a mother who was herself rapidly losing her sight. She was physically punished by her father if she made any sounds. Most of the time she was kept harnessed into an infant's potty chair; otherwise she was confined in a homemade sleeping bag in an infant's crib covered with wire mesh. She was fed only infant food.

The details of her discovery are not pertinent to this discussion, nor is speculation concerning the psychotic reasons behind the parents' actions. We have little information on the nature and extent of Genie's linguistic input during her isolation. The father's intolerance of noise is known, and there was no television or radio in the home. The periods of any human contact during the day were extremely limited. We know that her father and older brother did not speak to her, but at times barked at her like dogs. For the most part, hour after hour, day after day, year after year she was alone and constrained in her prison.



*Curtiss et al. (1974)*

# What did Genie learn?

During the four years that her language was observed, Genie learned quite a bit:

- comprehension of hundreds of words
- comprehension of some aspects of morphology (particularly good with plurals)
- comprehension of some prepositions

(16) PREPOSITIONS II: *under*, *over*, *in*, *on*, *behind*, *in front of*. Buttons and plastic boxes of different colors and sizes were used. Genie's task was to manipulate the buttons and boxes in accordance with the instructions.

Sample item: Put the red box behind the blue box.

Results 10/72–8/73:

| | Correct | Incorrect |
|---|---|---|
| *on* | 36 | 9 |
| *in* | 37 | 4 |
| *over* | 11 | 2 |
| *under* | 17 | 7 |
| *in front of* | 20 | 6 |
| *in back of* | 7 | 3 |
| *next to* | 7 | 1 |
| *behind* | 12 | 5 |

# What *didn't* Genie learn?

Genie's production lagged severely behind her comprehension:

- some lag is typical of development
- remember that she was punished for making sounds as a child

Her grammatical abilities developed much more slowly than her vocabulary.

(2) CONJUNCTION. Five familiar objects were placed in a row in front of Genie. She was asked to point to one or more of them in response to the test sentences.

Sample items: Show me the fork and pencil.
Show me the crayon or the knife.
Show me either the spoon or the crayon.

Results: Early responses to *or* conjunction treated same as *and* sentences. Later responses to *or* showed differentiation between *and* and *or* but non-comprehension of meaning of (*either*)/*or*.

| 10/71–12/73 | Correct | Incorrect | Unanalysable responses |
|---|---|---|---|
| *and* sentences | 100 | 3 | 0 |
| *or* sentences | 7 | 51 | 6 |

## Lots of questions:

Was Genie learning disabled?

What were the residual impacts of neglect and abuse on her social development?

…etc.

BUT:

After four years, funding to study her was pulled, so we can't know what she might have learned after a longer time.

Nonetheless, the disparity between her vocabulary development and grammatical development is widely interpreted as suggestive that grammar learning in a first language is specifically impaired after a sensitive period.

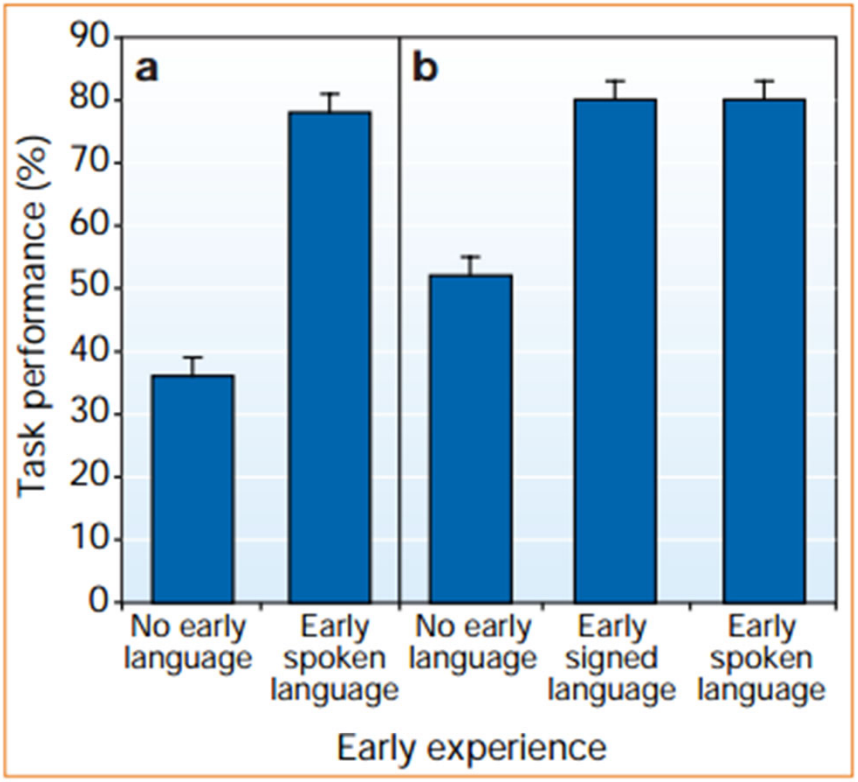



**Figure 1** Effects of early experience on later language learning. **a,** American Sign Language (ASL) performance of deaf adults who had experienced no language in early life, and of deaf adults who had experienced spoken language in early life. Subjects were tested using a task requiring recall of complex ASL sentences. **b,** English performance of deaf adults who had had no experience of language in early life, of deaf adults who had experienced ASL in infancy, and of hearing adults who had experienced a spoken language other than English in infancy. Subjects were tested using a task requiring judgements of whether complex English sentences given in print were grammatically correct; chance performance is 50%. Further details are available from the authors.

Many deaf children do not get sign language input, but have an otherwise typical upbringing.

Children (ages 9-15) have difficulty learning American Sign Language (ASL) if this is their first time learning a language.

A comparison group of children who went deaf after initially learning a spoken language do much better.

These differences persist into adulthood.

*Mayberry, Lock, & Kazmi (2002, Nature)*

# So, is there a “critical period” or a “sensitive period” for L1?

- Recovery from stroke, and certainly hemispherectomy looks very, very different in childhood vs. adulthood.

- Deprivation of input does not appear to extend the critical period.

- All of these data are from highly unusual cases with lots of other things going on. Difficult to reach firm conclusions. (Also note lack of animal models; can’t do gene knockout experiments with people…)

For next session:

Chapter 9

Nova doc on Genie: http://www.pbs.org/wgbh/nova/education/programs/2112_wildchil.html

Read in-depth description of Genie: http://ling.umd.edu/~omaki/teaching/Ling240_Summ2007/Curtiss1974_genie.pdf

The Enigma of Kaspar Hauser: http://www.imdb.com/title/tt0071691/

Valproate and perfect pitch: http://www.livescience.com/42347-valproate-perfect-pitch-absolute-pitch.html

Poto and Cabengo: http://www.criterion.com/films/27948-poto-and-cabengo